

No.	Answer	Solution
12	C	Initial total kinetic energy of the gas molecules $U_i = \frac{3}{2}NkT = \frac{3}{2}pV = \frac{3}{2} \times 1.0 \times 10^5 \times 0.010$ $= 1500 \text{ J}$ Since the volume of the change is changed to 0.0050 m^3 at constant temperature, there is no change in internal energy of the ideal gas. Total final kinetic energy of the gas molecules $U_f = 1500 \text{ J}$
13	D	Loss in kinetic energy of pellet = Gain in thermal energy of pellet